VIJAY KUMAR S

Aspiring Machine Learning Engineer | Python | PyTorch | GenAI (RAG, LLMs)

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SUMMARY

Aspiring Machine Learning Engineer with hands-on projects in time-series forecasting (PyTorch RNN) and RAG chatbots (LLMs, MCP), built end to end with Python, FastAPI and Streamlit. 1 year at Amazon as a Manual Tester, including data checks with SQL. MBA (Finance and Marketing) and B.E. in Mechanical Engineering. Seeking an entry-level Machine Learning Engineer / Data Scientist role.

SKILLS

Machine Learning & Deep Learning: Scikit-learn, PyTorch (RNN), Time-Series Forecasting, Feature Engineering, Model Evaluation (MAE, RMSE, R^2), NumPy, Pandas, Matplotlib, Seaborn

GenAI: LLMs (Groq API), RAG, MCP (Model Context Protocol), Prompt Engineering, Chatbots

Programming: Python, SQL (Joins, Sub-queries, Functions), Java (OOP, Data Structures)

Backend & Tools: FastAPI, Streamlit, SQLite, JavaScript, Chart.js, HTML/CSS, Jupyter Notebook, VS Code

Testing: Manual Testing, SDLC / STLC, Test Case Design

PROJECTS

PowerPulse: Electricity Demand Forecasting Web App github.com/Vijaykumar1159/PowerPulse

Python, PyTorch, Scikit-learn, Pandas, NumPy, FastAPI, JavaScript, Chart.js, HTML/CSS

- Built an end-to-end system to forecast hourly electricity demand (MW) from demand, temperature, hour-of-day and day-of-week features.
- Trained a PyTorch RNN on 24-hour sliding windows with a chronological 80/20 train/test split and StandardScaler; evaluated with MAE, RMSE and R^2.
- Implemented recursive multi-step forecasting for 1 to 168 hours ahead, with user-set temperature limits.
- Developed a FastAPI endpoint for CSV upload and a Chart.js frontend with demand charts and a forecast table.

Interview Prep Chatbot (RAG + MCP + Groq)

Python, Groq LLM API, RAG, MCP, SQLite, Streamlit

- Built a RAG chatbot that generates interview questions and explains topics from personal study notes, using LLMs served via the Groq API.
- Created MCP servers for progress tracking (SQLite), web / docs lookup and notes access.
- Designed a Streamlit app with Quiz, Explain, Revise and Dashboard modes, MCQ and chat practice, a score tracker and a weak-topics chart.

EXPERIENCE

Manual Tester | Amazon 08/2025 – Present

- Wrote and executed test cases across the SDLC / STLC; ran smoke, sanity and black-box testing on new builds.
- Logged and tracked defects with clear steps to reproduce, and re-tested fixes before release.
- Used SQL queries to validate data in the database during testing.

EDUCATION

MBA (Finance and Marketing) | Anna University Regional Campus Madurai 08/2021 – 05/2023

B.E. Mechanical Engineering | Velammal College of Engineering and Technology 08/2017 – 05/2021

HSC (12th): 90.8% (2017) | **SSLC (10th):** 97.8% (2015) | Parivallal Matric Higher Secondary School

CERTIFICATIONS

- Java, SQL and Manual Testing, QSpiders
- Data Science & AI Course: Python, SQL, Pandas, Machine Learning, NLP, GenAI (RAG, LLMs, MCP, Agentic AI)

Languages: Tamil, English